- Kidneys also produce erythropoietin, which is essential for the synthesis of RBC.
- Renin is released from the kidneys in response to low BP directly and also stimulates the production of aldosterone.
- Parathyroid hormones, calcitonin, insulin and gastrin are degraded by the kidneys.

10.2. TYPES OF KIDNEY DISEASES

The following are the types of kidney diseases:

- **Glomerulonephritis:** This is an inflammation of the glomeruli. Glomerulonephritis can be caused by infections, drugs or congenital abnormalities.
- **Kidney stones:** Kidney stones are another common kidney problem. Passing kidney stones can be extremely painful, but they rarely cause significant problems.
- **Chronic kidney disease:** The most common form of kidney disease is chronic kidney disease. Chronic kidney disease is a long-term condition that doesn't improve over time. It is commonly caused by high blood pressure. Diabetes is also a major cause of chronic kidney disease.
- **Polycystic kidney disease:** Polycystic kidney disease is a genetic disorder that causes numerous cysts (small sacs of fluid) to grow in the kidneys. These cysts can interfere with kidney function and cause kidney failure.
- **Urinary tract infections:** Urinary tract infections (UTIs) are bacterial infections of any part of the urinary system. Infections in the bladder and urethra are the most common. They are easily treatable and rarely lead to more health problems. However, if left untreated, these infections can spread to the kidneys and cause kidney failure.

Glomerular Filtration Rate (GFR)

GRF is a test used to check how well the kidneys are working. Specifically, it estimates how much blood passes through each unit of glomeruli. A GFR of 60 or higher is in the normal range. A GFR of 15 or lower may mean kidney failure.

10.3 GLOMERULONEPHRITIS

It is an inflammatory process affecting the glomeruli, the small blood vessels in the head of the nephron. It is most common in its acute form in children 3 to 10 years of age although it can also occur in adults past age 50.

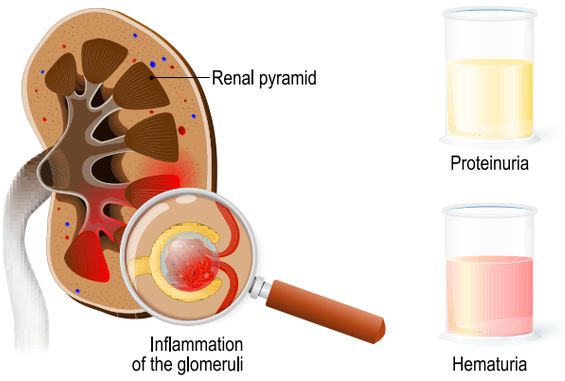


Fig 10.1 GLOMERULONEPHRITIS

10.3.1. Causes

The most common cause of glomerulonephritis is streptococcal infection. Renal infarction, acute pyelonephritis and metallic poisoning also cause glomerulonephritis. Other causes include primary kidney diseases such as IgA nephropathy and hereditary nephritis. Secondary nephritis can occur due to Systematic Lupus erythromatous (SLE).

Blood Urea Nitrogen (BUN)

BUN can provide important information about the kidney function. The test measures the amount of urea nitrogen in the blood. In general, around 7-20 mg/dl is considered as normal. High level BUN is a sign that the kidneys are not working.

Unit 10 Diet in kidney diseases Final 12-12-2018.indd 162 25-02-2019 15:37:52